Q.P. Code: 36220

Third Semester B.B.A. Degree Examination
January/February 2024
(NEP Scheme)
Statistics for Business Decisions

Time : 2 Hours Max. Marks : 60

Answer all the following questions.

1. a) What is Statistics? **(02)**

   b) Mention the types of Classification. **(04)**

   c) Draft a Blank table to show the distribution of any employee in an office according to sex, salary grade (Below Rs. 3,000, 3,000 to 5,000, 5,000 & above). Designation (Supervisor, Assistant and Clerk). **(06)**

OR

   d) Represent the following data prepare sub-divided percentage bar diagram from the data given below: **(12)**

Item of expenses	Food	Cloth	Education	Rent	Others
Family A	1350	1000	1800	850	600
Family B	1200	950	1600	850	800

2. a) Give the meaning of Mode. **(02)**

   b) Find the Median.

   360, 320, 280, 220, 260, 200, 180, 400 **(04)**

   c) Find out the standard deviation for the following data. **(06)**

Production in tons	12	14	16	18	20	22	24
No. of factories	6	12	18	26	16	10	8

OR

   d) From the following data calculate Mean, Median and Mode. **(12)**

C-I	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Frequency	4	7	15	11	6	9	4	3

3. a) What is Correlation. **(02)**

   b) Mention the different types of Correlations. **(04)**

   c) From the following data obtain two regression equations. **(06)**

Particulars	X	Y
Arithmetic Mean	80	110
Standard Deviation	4	5

Correlation co-efficient 0.7.

OR

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d) Calculate Karl Pearson's co-efficient of correlation and its probable error from the following : **(12)**

Price	8	10	15	17	20	22	24	25
Supply	25	30	32	35	37	40	42	45

4. a) What is Time series. **(02)**

b) Write a note on Least Square Method. **(04)**

c) Explain the components of Time Series. **(06)**

OR

d) Fit a straight line trend to these values and plot the trend line along with original data on a graph. **(12)**

Year	2002	2003	2004	2005	2006	2007	2008	2009	2010
Production	50	42	65	72	87	53	61	94	94

Estimate the production for the year 2011.

5. a) What is Index Number? **(02)**

b) Write the steps in the construction of Index Number. **(04)**

c) Calculate Index Number by using family budget method. **(06)**

Items	Weights	Price	
		2020	2022
Food	10	150	225
HR	5	150	150
Fuel	2	30	60
Clothing	3	30	75
Miscellaneous	5	50	75

OR

d) From the data given below: **(12)**

(1) Construct Fisher's Ideal Index Number.

(2) Verify how it satisfies TRT and FRT.

Commodity	2018		2019	
	Price	Quantity	Price	Quantity
Rice	4	40	5	50
Wheat	8	64	9	80
Jowar	10	70	10	70
Fuel	2	10	4	16

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